WEEK 04

The Browser(s): Your Window to the Web

Web Applications Development · Front-End Development Track

Prepared by Cyber.SoHo

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Agenda

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Learning Objectives

- Explain what a browser does behind the scenes when you visit a website
- Describe the rendering pipeline — from raw HTML to pixels on screen
- Manipulate the DOM (Document Object Model) with JavaScript
- Navigate Browser Developer Tools with confidence
- Recognize why the same site looks different across browsers
- Distinguish cookies, localStorage, and sessionStorage
- Read an HTTP request/response cycle in the Network tab
- Understand the Same-Origin Policy and why browsers enforce it

SECTION 01

What Even IS a Browser?

⏱️ ~8 minutes

"Most people think a browser is the app where you type google.com.

That's like saying a car is the thing with a steering wheel."

A Browser Has Six Jobs

📡

Send Requests

Via HTTP/HTTPS to web servers

📦

Receive Responses

HTML, CSS, JS, images, fonts

📋

Parse

Turn files into structured data (DOM, CSSOM)

🖥️

Render

Draw the visual result on screen

⚡

Execute JavaScript

Make the page interactive

🔐

Manage Security

Isolate tabs from each other

The Big Five Browsers

Browser	Rendering Engine	JS Engine	Maintained By
Chrome	Blink	V8	Google
Firefox	Gecko	SpiderMonkey	Mozilla Foundation
Safari	WebKit	JavaScriptCore	Apple
Edge	Blink (since 2020)	V8	Microsoft
Opera	Blink	V8	Opera Software

Three Engines Rule the Web

Blink

Chrome, Edge, Opera, Brave, Vivaldi

If it works in Chrome → likely works in Edge & Opera

Gecko

Firefox

Standalone engine — may interpret CSS/JS differently

WebKit

Safari + ALL iOS browsers

Apple requires WebKit on iOS — even Chrome on iPhone uses it

Analogy: The Pizza Delivery 🍕

Step	Pizza Equivalent	Browser Equivalent
1. Order	You call the pizza place	User types a URL
2. Connect	Phone line connects	DNS + TCP/TLS connection
3. Prepare	Kitchen makes your order	Server builds response
4. Parse	Driver reads the address	Browser parses HTML/CSS
5. Deliver	Pizza on your plate	Browser renders pixels
6. Interact	You add toppings	JavaScript runs

Section 1 Summary

🔑 **KEY TAKEAWAY**

A browser is not a passive window — it is a request engine, parser, renderer, script executor, and security guard all in one application. Understanding engines (Blink, Gecko, WebKit) lets you predict cross-browser behavior before it bites you.

SECTION 02

The Rendering Pipeline

⏱️ ~10 minutes

8 stages — from the moment you hit Enter to pixels on your screen.

Think of it as an assembly line in a factory.

From URL to Pixels: 8 Stages

1	2	3	4	5	6	7	8
DNS Lookup	**TCP/TLS**	**HTTP Req/Res**	**HTML → DOM**	**CSS → CSSOM**	**Render Tree**	**Layout**	**Paint**
Domain → IP address	Secure connection	Request files	Parse structure	Parse styles	Merge visible	Calculate positions	Draw pixels

Stage 1–3: Network Phase

- **DNS Lookup:** Converts human-readable domain name (github.com) to machine-readable IP address (140.82.121.3). Internet's phone book, coordinated by ICANN.
- **TCP/TLS Connection:** Opens a reliable connection. TLS encrypts HTTPS traffic — like sealing your letter in an envelope before mailing.
- **HTTP Request/Response:** Browser says "Give me /index.html" → Server responds with the file and a status code (200 = OK, 404 = Not Found).

Stage 4–5: Parsing Phase

HTML → DOM

- Reads HTML character by character
- Builds tokens (opening tags, closing tags, text)
- **Assembles into a tree structure:** the DOM
- Every tag becomes a node object

CSS → CSSOM

- Reads all CSS rules simultaneously
- Builds a parallel tree of style rules
- Maps selectors to visual properties
- Processed independently from HTML

Stage 6–8: Rendering Phase

- **Render Tree:** Merges DOM + CSSOM → only visible elements remain. Elements with display:none are excluded.
- **Layout (Reflow):** Calculates exact position and size of every element. "This div is 400px wide, starts 120px from the top."
- **Paint & Composite:** Fills in actual pixels — colors, text, borders, shadows, images. Complex pages use GPU-accelerated layers.

🔑 **KEY TAKEAWAY**

Every page load follows the same 8-stage pipeline: DNS → TCP/TLS → HTTP → DOM → CSSOM → Render Tree → Layout → Paint. You can measure each stage using the browser's Performance API. Bottlenecks at any stage slow the entire page.

SECTION 03

The DOM (Document Object Model)

⏱️ ~10 minutes

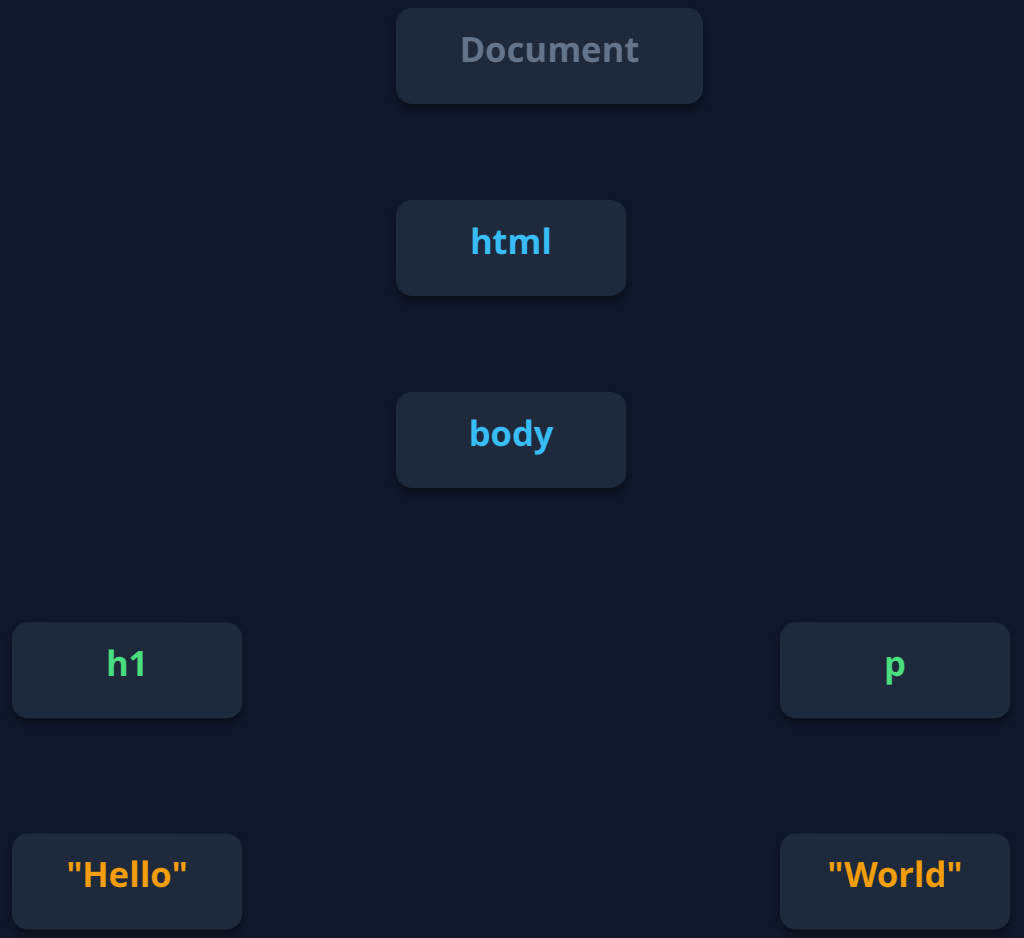
The single most important concept in front-end development.

What Is the DOM?

The DOM is the browser's live, in-memory representation of your HTML page as a tree of objects that JavaScript can read and modify.

- The DOM is NOT your HTML file — it's a living data structure built from parsing HTML
- JavaScript changes the DOM → Browser re-renders only what changed
- This is how modern apps feel "alive" without full page reloads
- Maintained by the W3C and WHATWG (living standard at dom.spec.whatwg.org)

DOM Tree Structure



Relationships:

- **Parent → Child:** body contains h1
- **Siblings:** h1 and p are siblings
- **Text Nodes:** leaf-level data

Core DOM Methods

Method	Purpose
document.getElementById()	Find element by its unique ID
document.querySelector()	Find first match by CSS selector
document.createElement()	Create a new DOM node in memory
parent.appendChild(child)	Attach a child node to a parent
parent.removeChild(child)	Detach a child node from the tree
element.textContent	Read or write text inside an element
element.innerHTML	Read or write HTML inside an element

🔑 **KEY TAKEAWAY**

The DOM is a tree of objects, not a text file. JavaScript manipulates the DOM → the browser re-renders only what changed. Master createElement, appendChild, removeChild, and querySelector — they are the foundation of every front-end framework.

SECTION 04

Browser DevTools

⏱️ ~8 minutes

Your X-Ray vision into any website.
Built into every modern browser.
No installation needed.

Opening DevTools

Operating System	Keyboard Shortcut
Windows / Linux	F12 or Ctrl + Shift + I
macOS	Cmd + Option + I

You can also right-click any element and choose "Inspect"

The Six Essential Panels

Elements

Inspect & edit live DOM and CSS in real time

Console

Run JS commands, view errors, warnings, and logs

Network

Watch every HTTP request the browser makes

Sources

View & debug JavaScript files with breakpoints

Application

Inspect cookies, localStorage, sessionStorage, caches

Performance

Record & analyze rendering speed and bottlenecks

Console Panel Essentials

Command	Output
console.log()	Normal informational message
console.warn()	Yellow warning icon
console.error()	Red error icon
console.table()	Tabular data in a formatted table
console.time() / timeEnd()	Measure execution time in ms

🔑 **KEY TAKEAWAY**

DevTools are not optional — they are the primary debugging tool for front-end development. A developer without DevTools is like a mechanic without a diagnostic scanner. Start with Elements, Console, and Network.

SECTION 05

Cross-Browser Compatibility

⏱️ ~8 minutes

Same recipe, different chefs, slightly different dishes.

Why Does the Same Page Look Different?

- **Different rendering engines:** Blink, Gecko, WebKit interpret specs with subtle differences
- **Feature support varies:** New CSS may work in Chrome but not in Safari for months
- **Default stylesheets differ:** Buttons, margins, and fonts look different per browser out of the box
- **JavaScript engine timing:** V8, SpiderMonkey, and JavaScriptCore ship new features at different dates

The Fix: CSS Resets & Tools

CSS Reset vs. Normalize

- **Reset:** Strips ALL defaults to zero
- **Normalize.css:** Preserves useful defaults, fixes inconsistencies
- Both ensure identical starting point across all browsers

Essential Tools

- **Can I Use:** Feature support per browser
- **MDN Web Docs:** Compatibility tables
- **Autoprefixer:** Adds vendor prefixes
- **BrowserStack:** Test on real devices

🔑 **KEY TAKEAWAY**

Cross-browser issues are inevitable, not bugs. Use Can I Use before adopting new features, apply a CSS reset or Normalize.css, and test in at least Chrome, Firefox, and Safari.

SECTION 06

Browser Storage

⏱️ ~7 minutes

Three ways your browser remembers things — each with different lifetimes, sizes, and visibility.

Three Client-Side Storage Options

Feature	🍪 Cookies	📦 localStorage	🔒 sessionStorage
Capacity	~4 KB	~5–10 MB	~5–10 MB
Lifetime	Until expiry date	Permanent	Until tab closes
Scope	Domain + path	Origin	Tab + origin
Sent to server?	YES (every request!)	NO	NO
API	Manual string parsing	setItem / getItem	setItem / getItem
Analogy	👛 Wallet	🧳 Suitcase	🔒 Hotel Safe

Origins & When to Use Each

Origin = Protocol + Domain + Port. https://example.com:443 is a DIFFERENT origin from http://example.com:80 — storage is isolated between them.

- **Cookies:** Auth tokens, server-side session tracking, preferences the server needs to read
- **localStorage:** Client-side caching, theme preferences, saved form drafts, offline data
- **sessionStorage:** Temporary wizard/form state, data that should die when the tab closes

🔑 **KEY TAKEAWAY**

Cookies go to the server with every request (keep them small!). localStorage persists forever (great for preferences). sessionStorage dies with the tab. Never store passwords or sensitive data in any client-side storage.

SECTION 07

HTTP & the Network Tab

⏱️ ~6 minutes

The language browsers and servers speak to each other.

HTTP Methods

Method	Purpose	Example
GET	Retrieve data	Loading a web page
POST	Send data to server	Submitting a login form
PUT	Update existing resource	Editing a user profile
DELETE	Remove a resource	Deleting a comment
PATCH	Partial update	Changing one field

HTTP Status Codes (1/2)

Code	Name	Meaning
200	OK	Request succeeded — here's your content
201	Created	New resource successfully created
301	Moved Permanently	URL has changed — update bookmarks
304	Not Modified	Cached version still valid — no download needed

🟢 2xx = Success 🟡 3xx = Redirection

HTTP Status Codes (2/2)

Code	Name	Meaning
400	Bad Request	Malformed syntax — check your request
401	Unauthorized	Login required — provide credentials
403	Forbidden	Authenticated but no permission
404	Not Found	URL doesn't exist — check for typos
429	Too Many Requests	Rate limited — slow down
500	Internal Server Error	Server bug — not your fault
503	Service Unavailable	Server overloaded or in maintenance

🟠 4xx = Client Error (your fault) 🔴 5xx = Server Error (their fault)

🔑 **KEY TAKEAWAY**

HTTP is the language browsers and servers speak. Every resource is a separate request with a method (GET/POST), a status code (200/404/500), and a body. The Network tab is the first place to look when something doesn't load.

SECTION 08

Browser Security Basics

⏱️ ~3 minutes

Browsers are under constant attack.
Four layers protect you.

Four Layers of Browser Security

HTTPS

All data encrypted via TLS between browser ↔️ server. Free certs from Let's Encrypt.

Same-Origin Policy

Scripts from evil.com CANNOT read data from bank.com. Origin = protocol + domain + port.

CORS

Controlled exception to SOP. Server explicitly grants cross-origin access via headers.

CSP

Server tells browser: "Only run scripts from these trusted sources." Blocks XSS attacks.

Analogy: The Apartment Building 🏢

Security Layer	Building Equivalent
HTTPS	Security cameras + encrypted mailboxes
Same-Origin Policy	Each apartment has its own lock
CORS	Tenant B gives Tenant A a spare key
CSP	Building posts list of approved delivery services

🔑 **KEY TAKEAWAY**

Browsers enforce security by default. Same-Origin Policy isolates every website. CORS creates controlled exceptions. CSP blocks unauthorized scripts. HTTPS encrypts everything in transit. These are the foundation of web security.

Week 04 Recap

Topic	One-Line Summary
Browser Role	Request engine + parser + renderer + JS executor + security guard
Rendering Pipeline	8 stages: DNS → TCP → HTTP → DOM → CSSOM → Render → Layout → Paint
DOM	Live tree of objects that JS can modify → triggers re-render
DevTools	F12 — Elements, Console, Network are your top 3 panels
Cross-Browser	Different engines = different results; use resets & Can I Use
Storage	Cookies (server), localStorage (permanent), sessionStorage (tab)
HTTP	GET/POST + status codes (200, 404, 500) = browser–server language
Security	SOP isolates, CORS allows exceptions, CSP blocks scripts, HTTPS encrypts

Essential Resources

- MDN Web Docs — developer.mozilla.org — Comprehensive web reference
- Can I Use — caniuse.com — Browser feature support tables
- Chrome DevTools — developer.chrome.com/docs/devtools — Official docs
- DOM Living Standard — dom.spec.whatwg.org — Official specification
- Normalize.css — necolas.github.io/normalize.css — CSS reset library
- HTTP Status Reference — developer.mozilla.org/en-US/docs/Web/HTTP/Status
- Let's Encrypt — letsencrypt.org — Free HTTPS certificates

COMING NEXT

Week 05: HTML & CSS Deep Dive

- Semantic HTML5 elements
- The CSS Box Model
- Flexbox and Grid layouts
- Responsive design with media queries

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